

Maurice Bastard

Curriculum Vitae

Aachen
Germany

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Education

- 04/2025–12/2026 **Master of Science, Computational Engineering Science,**
RWTH Aachen University, Aachen, Germany
Master's Thesis: In-Silico Shape Optimization of Catalyst Pellets Using Physics-Based Packed Bed Simulations and Machine Learning
- 10/2014–03/2025 **Bachelor of Science, Computational Engineering Science,**
RWTH Aachen University, Aachen, Germany
Bachelor's Thesis: Numerical Representation of Polydispersity Using EQMOM
- 08/2006–08/2014 **Abitur,**
German Higher Education Entrance Qualification, North Rhine-Westphalia, Germany

Professional Experience

- 06/2026–12/2026 **Graduate Research Assistant (Master's Thesis),**
IHE-3 – Forschungszentrum Jülich, Jülich, Germany
- Developed an automated simulation pipeline for the structural analysis of packed bed catalyst systems
 - Implemented parametric 3D modeling of pellet geometries (Python/Blender) and performed physics-based packed bed simulations using the Packed Bed Generator (PBG)
 - Engineered a Python interface for automated evaluation of geometric packed bed characteristics (porosity, specific surface area) directly from STL files
 - Integrated simulations into a multi-objective optimization framework (e.g., NSGA-II/PyGMO) to identify Pareto-optimal pellet morphologies
 - Maintained software repositories and automated computational and post-processing workflows on Linux HPC clusters
- 01/2023–08/2026 **Research Assistant,**
FVT – Chair of Process Engineering, RWTH Aachen University, Aachen, Germany
- Designed, conducted, and documented laboratory experiments on a multiphase loop reactor to characterize fluid flow behavior
 - Developed CFD simulations to model hydrodynamic reactor behavior
 - Built numerical evaluation tools in Python and MATLAB for experimental sensor data analysis and particle tracking
 - Onboarded, supervised, and mentored student research assistants
- 01/2022–12/2022 **Working Student – Production & Logistics,**
Saint-Gobain Vetrotex, Litomyšl, Czech Republic
- Analyzed and streamlined production and logistics processes to minimize downtime and increase operational efficiency
 - Coordinated and deployed continuous process improvements in collaboration with international cross-functional teams

- 06/2019–12/2021 **Working Student – Software Development & Analytics,**
Saint-Gobain Vetrotex, Aachen, Germany
- Developed automation software for sizing detection utilizing Visual Basic, C++, and Python
 - Performed competitive intelligence and market analyses to guide strategic positioning in an international industrial market
 - Organized and led global Microsoft Teams group training sessions and software onboarding workshops for sales teams
- 02/2018–06/2019 **Working Student – Event & Data Management,**
Saint-Gobain Adfors, Aachen, Germany
- Managed the organization and logistics of international conferences, trade fairs, and workshops across Germany, including booth design and support
 - Built internal databases, conducted data analysis, and created reporting dashboards; gained hands-on SAP experience
 - Fostered client relationships and networking to drive long-term business partnerships

Publications

- 2026 **Co-Author:** *Fluid dynamic characterization and operating window analysis of a pilot-scale multiphase loop reactor for biotechnological applications.*
 July 2026 – Chemical Engineering Research and Design.
DOI: doi.org/10.1016/j.cherd.2026.06.054

Languages

Language Skills German (Native), French (Native), English (Full Professional Proficiency)

Technical Skills

HPC & Parallel Computing **Parallelization:** MPI, OpenMP, CUDA, OpenACC
Numerics & Optimization: Eigen, NSGA-II, PyGMO

Simulation Frameworks **Physics & CFD:** OpenFOAM, LAMMPS, Modelica
Structural Mechanics: Finite Element Method (FEM)

Programming & Core **Languages:** C++, Python, Fortran, MATLAB, Java
Methods & Workflows: Agile/Scrum, Git, CMake, Linux/Unix, L^AT_EX

AI & Machine Learning **Frameworks & Models:** PyTorch, YOLO
AI-Assisted Development: Cursor, Claude, OpenCode

Visualization & Design Tools **Graphics & UI:** Qt, OpenGL
CAD & 3D: Blender, Allplan, Inventor, 3D Printing (CAD/CAM)
Data Analysis: Origin
Media Production: Adobe Photoshop, After Effects, GIMP

Interests & Personal Projects

- Leisure Rock climbing, skiing, traveling, photo & video production, tennis, badminton
- Projects & Coding Personal portfolio website
 Custom AI-powered smart home assistant
 Real-time hot air balloon tracking bot (Telegram API)
 Interactive 3D web application for visualizing ski touring and piste history